

GROUP 1

GREEN PATCH

DBSA ASSIGNMENT 1

GREEN PATCH

DESCRIPTION

GreenPatch is a platform that provides urban residents with limited gardening space to connect with property owners who have underutilized green areas.
The system facilitates **flexible agreements** for **renting**, **sharing**, or **co-cultivating** garden plots.



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- The system manages multiple **plots**, each identified by a unique plot ID. For each plot, the system stores its location and size. A particular owner owns each plot.

- Each **owner** has an Aadhar number, name, phone number, and residential address. An owner can possess multiple plots.



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- Plots are leased to **tenants** through rental agreements. A tenant is identified by a tenant ID (TID, which is also the Aadhaar number) and has associated personal information, including name, address, and phone number. All the tenants must lease at least one plot.

- The **agreement** captures the contractual relationship between an owner and a tenant. Each agreement includes the owner ID (OID), tenant ID (TID), plot ID (PID), lease start date, end date, size leased (in sq ft), and the price quoted per square foot.



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- The system manages gardening activities across multiple plots, each treated as a distinct unit of cultivation. Essential tasks such as soil preparation, planting, watering, fertilizing, weeding, pest control, and harvesting are stored in an **Activity table** (activity_id, activity_name).These are linked to specific plots through a **Gardening Schedule**, which records scheduled_date, completion_date, activity id, plot id, tenant id.

- This structure enables efficient tracking of work across plots, ensuring proper planning, resource allocation, and accountability in garden management.



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ENTITY AND ATTRIBUTES

PLOT

- Plot_id (Primary Key)
- Location
- Size
- Owner_id(Foreign key to Owner)

OWNER

- Owner_id (Primary Key)
- Name
- Address
- Phone Number

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ENTITY AND ATTRIBUTES

TENANT

- Tenant_id (Primary Key)
- Name
- Address
- Phone_number

AGREEMENT

- Owner_id (Foreign Key to Owner)
- Tenant_id (Foreign Key to Tenant)
- Plot_id (Foreign Key to Plot)
- Start date
- End date
- Price_quoted_per_square_feet
- Size

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ENTITY AND ATTRIBUTES

ACTIVITY

- Activity_id
- Activity_type

GARDENING SCHEDULE

- Activity_id
- Plot_id
- Tenant id
- Scheduled date
- Completion date

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RELATIONSHIPS AND CARDINALITY





OWNS

- Description: An Owner owns a Plot.
- Cardinality: 1-to-many. An Owner can own multiple Plots, but each Plot is owned by only one Owner.
- Participation:
  Total Participation: Every Plot must be owned by an Owner. This is a total participation rule, indicated by a double line.

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RENTS

- Description: A Tenant rents a Plot.
- Cardinality: many-to-many. A Tenant can rent multiple Plots and Plot can be rented by multiple Tenant at a time.
- Participation:
    Total Participation: A Tenant must rent a Plot.
    Partial Participation: A Plot may or may not be rented by a Tenant.

PROVIDES

- Description: An Owner provides an Agreement.
- Cardinality: 1-to-many. An Owner can enter into many Agreements, but each Agreement is entered into by only one Owner.
- Participation:
    Partial Participation: An Owner may or may not enter into an Agreement.
    Total Participation: An Agreement must be associated with an Owner.



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LEASES

- Description: A Tenant Leases an Agreement.
- Cardinality: 1-to-many. A Tenant can sign many Agreements, but each Agreement is signed by only one Tenant.
- Participation:
  Total Participation: A Tenant needs to sign an Agreement.
  Total Participation: An Agreement must be signed by a Tenant. This is a total participation rule.

APPLIES TO

- Description: An Agreement applies to a Plot.
- Cardinality: 1 - to- Many. An owner may or may not enter into an agreement
- Participation:
  Total Participation: An Agreement must cover a Plot.
  Partial Participation: A Plot may or may not be covered by an Agreement.



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PERFORMS

- Description: A Tenant performs Gardening Activities
- Cardinality: many to many. A Tenant can perform multiple garden activities on different plots, and each activity performed by multiple Tenants
- Participation:
  Partial Participation: A Tenant may or may not perform gardening activity.
  Total Participation: An activity must be performed by a Tenant.

HAS

- Description: A Gardening schedule assigned to a plot
- Cardinality: m to n. Each plot can have a Gardening schedule and each Gardening schedule can be assigned to multiple plots.
- Participation:
  Total Participation: A Gardening schedule must be assigned to a plot.
  Partial Participation: A plot may or may not have a Gardening schedule.

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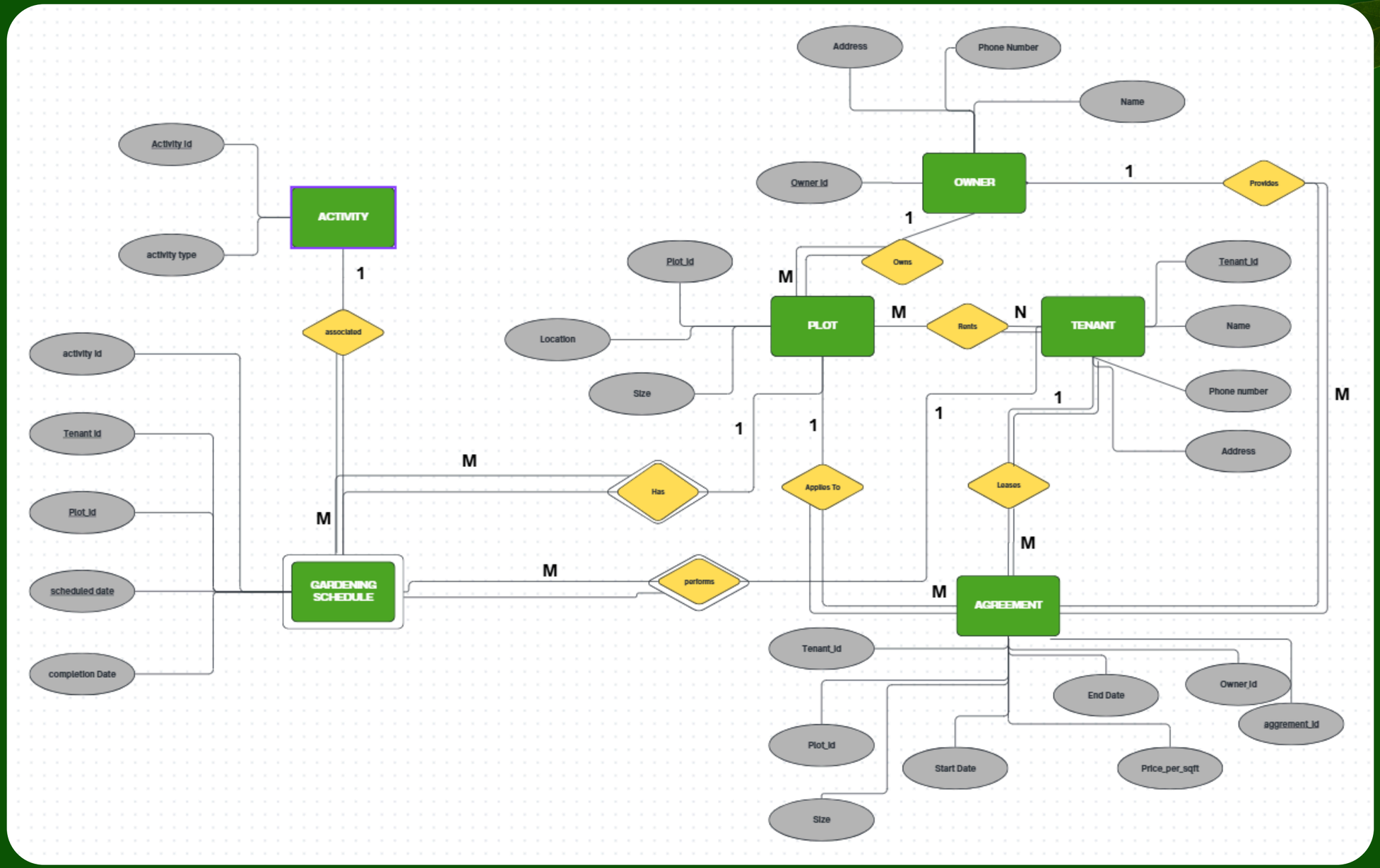
ASSOCIATED

- Description: A Gardening schedule associated with activities.
- Cardinality: 1 to many. One Gardening schedule associated with many activities, but each activity is associated with only on gardening a schedule.
- Participation:
    Partial Participation: A Gardening schedule may or may not include activities.
    Total Participation: An activity must be associated with a Gardening schedule.



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ER DIAGRAM



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OUR TEAM



Ajith B

2024MC19006



Deeksha R

2024MC19001



Sankarraj V

2024MC19016



Santhosh

2024MC19015



Sheetal

2024MC19005


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THANK YOU